Example 3.62 If $A=\begin{pmatrix}1 & 3 & -2\\ 5 & -4 & 6\\ -3 & 2 & 9\end{pmatrix}$, $B=\begin{pmatrix}1 & 8\\ 3 & 4\\ 9 & 6\end{pmatrix}$, find $A+B$.

Solution It is not possible to add A and B because they have different orders.

Multiplication of Matrix by a Scalar

We can multiply the elements of the given matrix A by a non-zero number k to obtain a new matrix kA whose elements are multiplied by k. The matrix kA is called scalar multiplication of A.

Thus if $A=(a_{ij})_{m\times n}$ then , $kA=(ka_{ij})_{m\times n}$ for all $i=1,2,\ldots,m$ and $\forall\ j=1,2,\ldots,n$.

Example 3.63 If $A=\begin{pmatrix}7 & 8 & 6\\ 1 & 3 & 9\\ -4 & 3 & -1\end{pmatrix}$, $B=\begin{pmatrix}4 & 11 & -3\\ -1 & 2 & 4\\ 7 & 5 & 0\end{pmatrix}$ then Find $2A+B$.

Solution Since A and B have same order 3×3, $2A+B$ is defined.

$$\text{We have } 2A+B = 2\begin{pmatrix}7 & 8 & 6\\ 1 & 3 & 9\\ -4 & 3 & -1\end{pmatrix}+\begin{pmatrix}4 & 11 & -3\\ -1 & 2 & 4\\ 7 & 5 & 0\end{pmatrix}=\begin{pmatrix}14 & 16 & 12\\ 2 & 6 & 18\\ -8 & 6 & -2\end{pmatrix}+\begin{pmatrix}4 & 11 & -3\\ -1 & 2 & 4\\ 7 & 5 & 0\end{pmatrix}$$

$$=\begin{pmatrix}18 & 27 & 9\\ 1 & 8 & 22\\ -1 & 11 & -2\end{pmatrix}$$

Example 3.64 If $A=\begin{pmatrix}5 & 4 & -2\\ \frac{1}{2} & \frac{3}{4} & \sqrt{2}\\ 1 & 9 & 4\end{pmatrix}$, $B=\begin{pmatrix}-7 & 4 & -3\\ \frac{1}{4} & \frac{7}{2} & 3\\ 5 & -6 & 9\end{pmatrix}$, find $4A-3B$.

Solution Since $A,\ B$ are of the same order 3×3, subtraction of $4A$ and $3B$ is defined.

$$4A-3B = 4\begin{pmatrix}5 & 4 & -2\\ \frac{1}{2} & \frac{3}{4} & \sqrt{2}\\ 1 & 9 & 4\end{pmatrix}-3\begin{pmatrix}-7 & 4 & -3\\ \frac{1}{4} & \frac{7}{2} & 3\\ 5 & -6 & 9\end{pmatrix}$$

$$=\begin{pmatrix}20 & 16 & -8\\ 2 & 3 & 4\sqrt{2}\\ 4 & 36 & 16\end{pmatrix}+\begin{pmatrix}21 & -12 & 9\\ -\frac{3}{4} & -\frac{21}{2} & -9\\ -15 & 18 & -27\end{pmatrix}$$

$$=\begin{pmatrix}41 & 4 & 1\\ \frac{5}{4} & -\frac{15}{2} & 4\sqrt{2}-9\\ -11 & 54 & -11\end{pmatrix}$$

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